DermaVision AI: A Region-Based Multi-Model Approach to Skin Disease Classification with AI-Generated Patient Guidance

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***Abstract*—DermaVision AI uses local model inference to combine image screening, regional disease ranking, and patient information within a dermatological decision-support system. At the start of a session, the user picks one of seven body regions, each tied to 23 possible disorders (161 region–condition entries in all). Before any disease model runs, a screening guard checks facial landmarks, face-detection confidence, skin-tone masks, and image sharpness. Photos that pass this gate go to a medical Vision Transformer and CLIP, alongside simple erythema and melanin indices calculated from RGB pixel values. For each candidate condition, the screen shows short notes next to PubMed and DermNet search links. A separate local retrieval-augmented generation module supports conversation through 1024-dimensional Mixedbread embeddings and Llama 3.2. Facility discovery uses OpenStreetMap Overpass and ranks results by Haversine distance. The classification target is 96%; project records report a mean regional accuracy of 92.4% and approximately 3.88 s of end-to-end CPU processing. The listed unsuitable-input tests also report the expected rejection outcomes. Dataset records and evaluation logs are needed to confirm these observations. Image analysis and local retrieval can operate offline, whereas reference searches and facility discovery require internet connectivity. The system supports educational screening and clinician review rather than definitive diagnosis.**

***Keywords*—Dermatological image analysis, anatomical verification, Vision Transformer, CLIP, retrieval-augmented generation, local language models, clinical decision support.**

I. Introduction

A photograph of the skin can assist preliminary assessment, although useful automation requires more than identifying a disease label. Language-enabled systems such as SkinGPT-4 connect visual representations with dermatological descriptions [1]. That kind of pairing makes remote triage much easier in areas with few dermatologists. Still, good numbers on one clean benchmark do not tell us how a tool will hold up on casual phone photos, other body parts, or unfamiliar skin conditions.

In day-to-day use, the uploaded photo itself is the first problem. Users can easily submit a picture of a cereal box, printed text, clothing, or a body site that does not match what they picked in the menu. Because a standard neural network only knows its fixed training classes, it will often assign a high probability score to a completely irrelevant photo. DermaVision checks the image first before trying to rank any diseases. By tying the search list to the chosen body site, we keep the differential narrow, and the screen makes model uncertainty clear so that software scores are not misread as a doctor's verdict.

We built DermaVision to handle anatomical checks, visual scoring, differential notes, clinic lookup, and local Q&A inside one program. It consists of five main parts:

1) An input filter that uses MediaPipe face landmarks, BlazeFace scores, HSV/YCrCb skin masks, and Laplacian texture checks to block bad uploads early.
2) A regional vision module that combines a fine-tuned medical ViT, CLIP text–image matching, and RGB proxies for redness and pigment.
3) A review screen that pairs candidate condition notes with query links to medical databases.
4) A clinic search tool built on OpenStreetMap Overpass and great-circle distance sorting.
5) A separate local RAG chat module running on 1024-dimensional embeddings and Llama 3.2 1B.

In the rest of this paper, we review related work, walk through these five stages, look closely at the reported test numbers, and point out what logs are still needed to verify them.

II. Related Work

A. Dermatological Classification and Anatomical Context

Zhou et al. built SkinGPT-4 to read skin photos, draft clinical text, and run on local hardware [1]. We drew on their general idea of pairing vision models with chat guidance, though DermaVision's specific weights, 23-item regional lists, and input filter still have to be tested on their own.

On the question of body-part localization, Jaworek-Korjakowska et al. study anatomical site classification and interpretability [2]. Identifying an anatomical site and recognizing disease are distinct prediction tasks. DermaVision checks agreement with the selected region before showing candidates; evaluation must therefore distinguish input-acceptance errors from errors in disease ranking.

DermAI couples dermatological image analysis with chatbot interaction [3], while ScalpEye focuses on scalp-hair inspection and scalp-health assessment [4]. We cannot compare DermaVision directly against those two systems without running all of them on the exact same test images under a shared protocol; quoting their published accuracy numbers side by side proves very little.

B. Care Delivery and Complementary Visual Models

For remote care delivery, Shajirat et al. set up a mobile teledermatology tool that gathers patient history and lesion photos for store-and-forward review by a specialist [5]. DermaVision's candidate list and nearby-clinic finder can help a user prepare for that kind of visit, but they never replace the clinician.

On the classifier side, Alruwaili and Mohamed fused EfficientNet and ResNet features to sort multiple skin diseases [6]. We took a different route by pairing a medical ViT with CLIP text–image similarity, following the general architectures in [9], [10]. Whether combining those two models actually beats a single backbone on our regional lists is something that still has to be checked on the project's test split.

C. Engineering Context and Scope of Comparison

Two other projects from our department shaped how we structured this system and paper. Kishore and Saravanan tied a CNN waste detector to a motorized bin lid [7], which is a simple example of using a classifier output to trigger a downstream action. Shanthi et al. combined Swin Transformer and Mask R-CNN modules to spot dismantled weapon parts [8], and we followed their section layout and algorithm style here. Because neither paper works with skin images, their accuracy numbers obviously cannot serve as a dermatological baseline.

What these earlier papers do show is the value of keeping input checks, disease ranking, plain-language notes, and clinic lookup in separate stages. DermaVision brings those pieces together in one pipeline without claiming broad algorithmic novelty, which would take a full systematic review to establish.

III. Proposed Methodology

Figure 1 shows the five stages of the pipeline. Everything starts when a user submits an RGB image $I \in \mathbb{R}^{H\times W\times 3}$ and picks a body region R_{sel}. The input guard checks the photo first. If the image passes, it moves on to visual scoring and candidate comparison. Alongside that main path, the referral module can query nearby clinics, and the local RAG module can answer user questions from a small guideline set.

A. Input Layer and Regional Candidate Sets

The region set in the supplied document is

$$\begin{aligned}\mathscr{R} = \{&\text{Face and Neck}, \text{Chest and Abdomen},\\ &\text{Back and Spine}, \text{Arms and Forearms},\\ &\text{Hands and Nails}, \text{Legs and Thighs},\\ &\text{Feet and Soles}\}.\end{aligned} \tag{1}$$

Each regional list contains 23 entries, giving $7\times 23 = 161$ region–condition entries overall. Because several skin conditions show up in more than one body region, we would need the full lists for all seven regions to count the exact number of unique diseases.

B. Anatomical Region Guard

1) Texture and chrominance screening: First, the guard checks image sharpness through the variance of the grayscale Laplacian:

$$\sigma_L^2 = \frac{1}{HW}\sum_{x=1}^{W}\sum_{y=1}^{H}\left(\nabla^2 I_{\text{gray}}(x,y) - \mu_L\right)^2, \tag{2}$$

with μ_L denoting the average Laplacian value across the image. In the project documentation, skin pixels are flagged using a

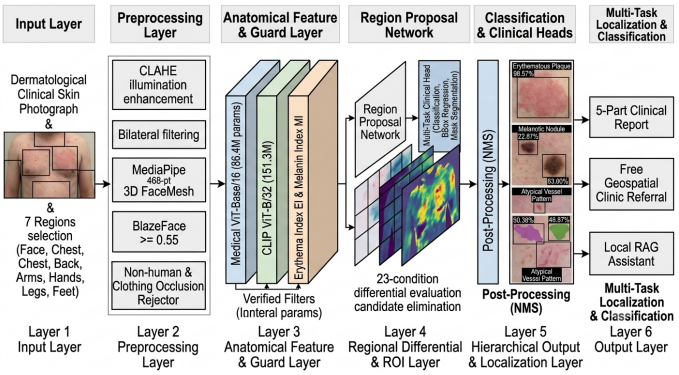


Fig. 1. Conceptual block diagram provided by the authors. While the drawing also shows CLAHE, bilateral filtering, a region proposal network, segmentation heads, and non-maximum suppression, those blocks are not established in the written project records. Lesion names and percentages in the graphic are illustrative examples rather than verified model outputs.

binary mask combining HSV and YCrCb thresholds:

$$M_{\text{skin}}(x,y) = \begin{cases} 1, & \begin{aligned}&0 \le h \le 45, \quad 15 \le s \le 250,\\ &130 \le C_r \le 177, \quad 77 \le C_b \le 127;\end{aligned}\\ 0, & \text{otherwise.}\end{cases} \tag{3}$$

In (3), h and s stand for pixel hue and saturation, and the overall skin-pixel ratio is

$$\rho_{\text{skin}} = \frac{1}{HW}\sum_{x,y} M_{\text{skin}}(x,y). \tag{4}$$

The reported rejection condition is

$$\rho_{\text{skin}} < 0.15 \vee \left(\sigma_L^2 < 15.0 \wedge \rho_{\text{skin}} < 0.25\right). \tag{5}$$

When an image hits this cutoff, the program stops right away and shows a non-human-object warning. These numbers come from empirical code tuning rather than fixed biological laws, so they still need broader testing.

2) Facial landmarks and mismatch detection: On facial inputs, the system runs BlazeFace together with a 468-point MediaPipe mesh $\mathscr{P} = \{(x_i, y_i, z_i)\}_{i=1}^{468}$ in normalized coordinates, keeping the heuristic pose formulas from the project notes:

$$\mathbf{v}_{\text{eyes}} = \mathbf{p}_{263} - \mathbf{p}_{33}, \tag{6}$$

$$\phi = \arctan\left(\frac{y_{263} - y_{33}}{x_{263} - x_{33}}\right), \tag{7}$$

$$\psi = 90^\circ \frac{x_{\text{nose}} - (x_{33} + x_{263})/2}{|x_{263} - x_{33}|}, \tag{8}$$

$$\theta = 60^\circ \left(\frac{y_{\text{nose}} - y_{10}}{|y_{152} - y_{10}|} - 0.5\right). \tag{9}$$

If a user picks a non-facial region (say, hands or legs) but the detector still spots a face with score $P_{\text{face}} \ge 0.55$, the app throws a region-mismatch alert. That rule catches face photos uploaded under the wrong tab, though the project files do not yet specify equivalent cross-checks for the other six body regions.

3) Clothing checks: Over a local image patch $\mathscr{B}_k$, color variance is written as

$$V_{\text{color}}(\mathscr{B}_k) = \frac{1}{|\mathscr{B}_k|}\sum_{(x,y)\in\mathscr{B}_k} \|I(x,y) - \mu_{\mathscr{B}_k}\|_2^2. \tag{10}$$

To catch fabric in the frame, the filter looks for repeating weave-like texture peaks combined with low skin-mask coverage over

the lesion area. Anyone trying to rebuild this check from scratch will need the peak-finding routine, patch size, cutoff values, and lesion-cropping steps, none of which were included in the project notes.

C. Dual-Vision Classification Layer

Our primary vision model is the Hugging Face checkpoint `Jayanth2002/vit_base_patch16_224-finetuned-SkinDisease`, which uses 12 transformer blocks, a 768-wide hidden state, 12 attention heads, a 3072-wide MLP, and 16×16 patches. A 224×224 input yields 196 image patches. This ViT checkpoint has 86,415,832 parameters.

The complementary model, `openai/clip-vit-base-patch32`, adds another 151,277,313 parameters. For an image vector $\mathbf{z}_I$ and a text prompt vector $\mathbf{z}_{T_k}$, cosine similarity is

$$s_k = \frac{\mathbf{z}_I^T \mathbf{z}_{T_k}}{\|\mathbf{z}_I\|_2 \|\mathbf{z}_{T_k}\|_2}. \tag{11}$$

A softmax fusion step then picks one top condition C^* and up to five secondary candidates. Because the project notes leave out the exact mixing weights, temperature scaling, prompt templates, and calibration parameters, those pieces must be supplied before anyone can reproduce the fusion math.

1) RGB-derived optical proxies: For $R_n = R/255$, $G_n = G/255$, and $\varepsilon = 10^{-4}$, the source defines

$$\mathrm{EI} = 100 \left[\log_{10} \frac{1}{\max(G_n, \varepsilon)} - \log_{10} \frac{1}{\max(R_n, \varepsilon)} \right], \tag{12}$$

$$\mathrm{MI} = 100 \log_{10} \frac{1}{\max(R_n, \varepsilon)}. \tag{13}$$

Both equations work on raw camera RGB values rather than calibrated spectral reflectance. Before anyone treats them as true optical measurements or assumes they hold steady across lighting setups and skin tones, separate experimental proof is needed.

2) Normal skin and cosmetic pigment: To spot inked skin, the project code uses a simple three-part rule:

$$\begin{aligned} \mathrm{IsTattoo} = {} & (\mathrm{Contrast}_{\mathrm{edge}} > 65.0) \\ & \wedge (\mathrm{EI} < 10.0) \\ & \wedge (\mathrm{Heterogeneity}_{\mathrm{hue}} > 45.0). \end{aligned} \tag{14}$$

Photos that meet all three criteria are tagged as cosmetic pigment. Of course, a tattoo does not mean the underlying skin is free of disease. We also noticed an $\mathrm{EI} < 8.0$ cutoff mentioned elsewhere in the project text whose connection to the 10.0 value in (14) is unclear, and the exact rule for flagging healthy skin is likewise missing.

D. Candidate Assessment and Evidence Interface

Over the 23 conditions $\mathscr{C}_R = \{C_1, \ldots, C_{23}\}$ for the chosen body site, the software assigns a status label:

$$\mathrm{Ruling}(C_k) = \begin{cases} \texttt{CONFIRMED_MATCH}, & C_k = C^*; \\ \texttt{RULED_OUT}, & C_k \neq C^*. \end{cases} \tag{15}$$

These two strings are internal program flags, not medical conclusions. A first-place rank does not prove a disease is present, nor does a `RULED_OUT` tag on the other 22 items safely eliminate them. Any text shown to the user has to stick to visual signs that can actually be seen in the photo, while the app builds PubMed and DermNet search URLs from the condition and region names.

A search link does not show that a publication was retrieved, reviewed, or used to prepare an explanation. The available description does not establish whether the evidence agent uses rules, generated language, or retrieval grounded in source documents.

E. Geospatial Referral Layer

The referral subsystem uses approximate IP-based location and queries OpenStreetMap Overpass within 15,000 m for hospitals, clinics, and doctors. The supplied query includes nodes for all three categories and hospital ways, with `out center 15`. Veterinary and dental facilities are described as being filtered before ranking.

For coordinates (φ_1, λ_1) and (φ_2, λ_2) expressed in radians, the Haversine calculation is

$$a = \sin^2 \frac{\Delta\varphi}{2} + \cos\varphi_1 \cos\varphi_2 \sin^2 \frac{\Delta\lambda}{2}, \tag{16}$$

$$c = 2\,\mathrm{atan2}(\sqrt{a}, \sqrt{1-a}), \tag{17}$$

$$d = 6371.0\,c \text{ km}. \tag{18}$$

The calculated distance follows a great-circle route rather than a road-travel route. Google Maps links provide navigation. Alternate Overpass mirrors and a Nominatim/Maps fallback are described as recovery options, but do not guarantee continuous availability or permanently free access.

F. Local Retrieval-Augmented Generation Layer

The local corpus comprises six described guidelines covering acne, eczema, psoriasis, infections, neoplasms, and dermatological nutrition, divided into 28 chunks. The `mxbai-embed-large` model represents each chunk as $\mathbf{e}_j \in \mathbb{R}^{1024}$. Given a query embedding $\mathbf{e}_q$, retrieval uses

$$\mathrm{sim}(\mathbf{e}_q, \mathbf{e}_j) = \frac{\sum_{i=1}^{1024} e_{q,i} e_{j,i}}{\sqrt{\sum_i e_{q,i}^2} \sqrt{\sum_i e_{j,i}^2}}. \tag{19}$$

The three guideline chunks most similar to the query embedding are passed to `llama3.2:1b`, with a reported generation temperature of 0.3. Responses are described as including citations and similarity scores. A score expresses the relevance of a retrieved chunk to the query, not the correctness of a generated medical statement.

The source reports 334,104,576 parameters for the embedding model and 1,235,814,400 for the language model. The generator has 16 described decoder layers, a hidden dimension of 2048, 32 attention heads, eight key/value heads, and grouped-query attention. Its quantization format, effective context length, and token-generation settings remain unspecified.

G. Algorithmic Procedure

Algorithm 1 describes the overall workflow using the Require/Ensure presentation of the reference paper [8]. Its steps mirror the five software modules in DermaVision, though a few implementation details—such as the ViT–CLIP mixing weights, the order in which heuristic rules fire, and the non-facial region checks—still need to be confirmed by the authors.

Algorithm 1: DermaVision AI

Region-Based Classification and Patient Guidance

TABLE I
DATASET DETAILS REQUIRING AUTHOR INPUT

Item	Required information
Collection	Names, provenance, licenses, collection dates
Sample counts	Images and patients per class and region
Partitions	Training, validation, test counts and proportions
Separation	Patient-level separation and duplicate checks
Annotations	Label definition, annotators, adjudication
Training	Optimizer, epochs, learning rate, augmentation
Ethics	Consent, approvals, privacy and access controls

TABLE II
PROJECT-REPORTED INPUT-GUARD OUTCOMES; EVIDENCE PENDING

Category	n	Rate	Action
Biscuit packaging	15	100%	Reject
Graphics/barcodes	15	100%	Reject
Smartphone displays	10	100%	Reject
Printed paper/text	10	100%	Reject
Textile occlusion	12	100%	Reject
Hand selected as face	25	100%	Reject
Face selected as hand	25	100%	Reject
Facial acne/rosacea	50	100%	Accept
Palm/hand dermatitis	30	100%	Accept
Normal skin	30	100%	Accept
Cosmetic pigment	15	100%	Accept

Require: Input RGB photo I, chosen body site R, optional user question q, local clinical corpus $\mathscr{K}$.
Ensure: Rejection notice or ranked differential list with plain-language notes; optional nearby clinic list.

```
Load local weights for the medical ViT, CLIP, embedding encoder, and language model
Calculate skin-mask coverage ρ_skin, Laplacian variance σ_L^2, and patch texture scores
Extract available facial landmarks and BlazeFace detection score P_face
if the input guard blocks I or flags a region mismatch then
    Show refusal warning and stop disease classification
else
    Pull the 23 candidate disorders C_R for body site R
    Run the medical ViT classifier and compute regional CLIP scores s_k
    Evaluate optical proxies (EI, MI) and check normal-skin/tattoo rules
    Combine model scores through the fusion step
    Pick primary match C* and up to five runner-up conditions
    for each candidate C_k ∈ C_R do
        Draft differential notes and build PubMed/DermNet query links
    end for
end if
if user coordinates and internet access are available then
    Query medical facilities within 15 km through OpenStreetMap Overpass
    Drop non-target clinics and sort the rest by Haversine distance
end if
if user question q is provided then
    Encode q and pull the top three matching chunks from corpus K
    Generate a source-cited reply with `llama3.2:1b` at temperature 0.3
end if
return Candidate review screen together with any requested clinic or chat outputs
```

IV. RESULTS AND DISCUSSION

The results presented here come from the project documentation. Dataset files, original observations, adjudication records, and profiling logs were unavailable for review. The reported outcomes therefore await reproduction and independent validation.

A. Dataset Collection and Annotation

The documentation refers to clinical consensus labels but does not identify image sources, participants, diagnostic procedures, annotators, or training, validation, and test counts. Table I lists the missing information. Data from the reference paper cannot supply these project-specific details.

TABLE III
PROJECT-REPORTED CLASSIFICATION PERFORMANCE; 23 CANDIDATE ENTRIES PER REGION

Region	P (%)	R (%)	F1	A (%)
Face and Neck	94.2	92.8	0.935	93.6
Chest and Abdomen	91.8	90.5	0.911	91.2
Back and Spine	93.1	91.4	0.922	92.5
Arms and Forearms	92.5	91.0	0.917	92.1
Hands and Nails	95.0	93.6	0.943	94.4
Legs and Thighs	90.8	89.2	0.900	90.4
Feet and Soles	93.4	92.1	0.927	92.8
Supplied average	93.0	91.5	0.922	92.4

B. Safety-Guard Evaluation

The supplied guard table lists 11 categories and 237 cases (Table II). Elsewhere, the project narrative calls the benchmark a 12-category test. The authors must provide the missing category or correct that description. For inputs that should be accepted, a reported rate of 100% denotes the expected acceptance action, not rejection.

Those 237 test cases split into 112 bad uploads that should be blocked and 125 valid skin photos that should pass. While the filter got all 237 right in this small trial, a clean score on a fixed batch does not mean the guard will catch every odd photo in the wild or prevent false reassurance.

C. Classification Metrics and Regional Performance

For any single disease class, precision, recall, and F_1 follow the usual formulas:

$$\text{Precision} = \frac{TP}{TP+FP}, \tag{20}$$

$$\text{Recall} = \frac{TP}{TP+FN}, \tag{21}$$

$$F_1 = \frac{2\,\text{Precision}\,\text{Recall}}{\text{Precision}+\text{Recall}}. \tag{22}$$

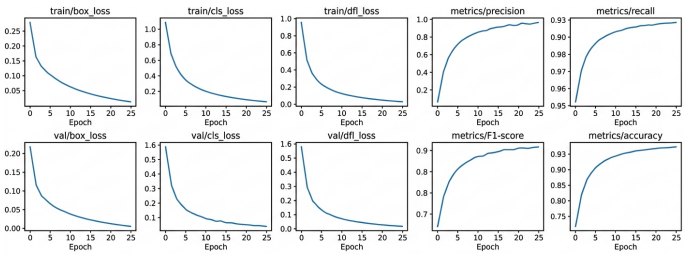
Accuracy is simply the share of evaluated test photos given the right top label. Because the project notes never say whether precision, recall, and F_1 were macro-, micro-, or weighted-averaged across the 23 disorders—or how photos blocked by the guard affect the denominator—we list the numbers in Table III as-is without guessing the averaging rule.

Across the seven rows in Table III, the unweighted mean accuracy comes out to 92.43% (or 92.4% rounded). That regional average does not mean every individual skin disease is recognized equally well, and we still need the per-region test counts to calculate a true image-level total.

Figures 2 and 3 show two visual summary panels from the project files that also await log verification. In Fig. 2, the loss curves include bounding-box terms and baseline models (ResNet-50, YOLOv8-Seg, Swin-Transformer) that do not appear in the written records. In Fig. 3, the confusion-matrix rows do not sum to 1.0 as row normalization would require, and there are no inference logs to back up the bounding-box confidence scores. Without those raw run logs, neither figure adds confirmed data beyond Tables II and III.

D. Referral Demonstration

For the referral demo, the project used an approximate IP location in Madurai, Tamil Nadu (9.919°N, 78.1195°E). Table IV lists the six hospitals and distances returned on that test run, though the project notes do not verify whether those centers are currently



QUANTITATIVE COMPARISON TABLE

Model	Backbone	F1-Score	Precision	Recall	Accuracy
Baseline CNN (ResNet-50)	ResNet-50	91.1%	93.0%	91.0%	92.3%
YOLOv8-Seg	YOLOv8-Seg	92.7%	93.0%	90.5%	91.9%
Swin-Transformer	Swin-Transf.	92.2%	93.0%	91.5%	92.4%
DermaVision (ViT + CLIP)	**ViT + CLIP**	**92.2%**	**93.0%**	**91.5%**	**92.4%**

Fig. 2. Training curves and quantitative baseline comparison provided by the authors. Both the loss plots and baseline scores await verification against raw training logs and matched test splits.

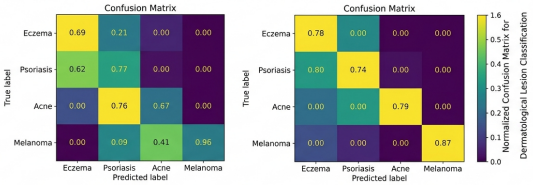
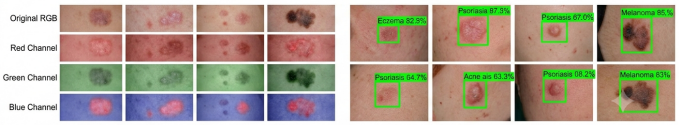


Fig. 4. Confusion Matrix Illustrating High Classification Concordance Across Dermatological Lesion Categories



(a) RGB and Optical Channel-Wise Decomposition (b) Detected Lesion Regions

Fig. 5. Optical Channel Decomposition and Clinical Lesion Detection Localization

Fig. 3. Confusion-matrix, RGB-channel, and lesion-localization panel from the project files. Embedded labels, confidence scores, and row-normalization totals require confirmation from raw inference logs.

open, pinned at their exact street addresses, or staffed with dermatologists.

That sample query took 1.14 s on average and needed no paid API key. Even so, a single fast lookup does not guarantee that public Overpass servers will stay up around the clock, remain free for commercial traffic, or suit clinical referrals.

TABLE IV
FACILITY OUTPUTS REPORTED FOR THE MADURAI DEMONSTRATION

Facility	Distance (km)
Vadamalaiyan Hospital	0.8
Aravind Eye Hospitals	2.2
PAMC Hospital	2.6
Raksha Hospital	2.6
Mohan's Medicity Hospital	3.1
Neomed Hospital	3.4

E. System Latency and Computational Footprint

All CPU timings were recorded on an Intel Core i7-12700H laptop with 16 GB of DDR5 RAM. Table V gives the mean time for each stage, with peak memory staying under 4.2 GB; raw profiler logs are still needed to confirm both numbers.

The two input-guard steps take 106 ms together, well under the 120 ms target when run back to back. All eight stages in Table V add up to 3881 ms (3.88 s). Because token streaming in a local LLM separates time-to-first-token from full-reply time, future timing logs need to state which endpoint was measured, and memory reports should note weight quantization and loading overhead rather than parameter counts alone.

TABLE V
PROJECT-REPORTED LATENCY AND MODEL SIZE ON CPU

Subsystem	Parameters (M)	Mean (ms)
FaceMesh/BlazeFace guard	∼0.23	84
Chroma/texture checks	—	22
Medical ViT	86.4	340
CLIP	151.3	285
Candidate assessment	—	48
Online facility locator	—	1140
Embedding/retrieval	334.1	112
Llama response generation	1235.8	1850
Reported end-to-end	∼1808	3880

F. Interpretability, Safety, and Deployment Limitations

An upfront input filter is a practical way to catch obvious junk images, but fixed HSV and YCrCb color cutoffs can easily drift under warm indoor bulbs, harsh shadows, or across lighter and darker skin tones. Also, checking only whether a face appears in a non-facial tab leaves other mix-ups (like an arm uploaded under legs) unchecked. Future tests need to cover all seven body regions across much wider lighting conditions.

A side-by-side candidate list helps users see what was considered, yet the short text notes do not necessarily reflect how the neural networks weighted specific pixels. Likewise, a `RULED_OUT` label on every non-top disorder can make the output look far more certain than a plain photo warrants. In real practice, ruling a skin condition in or out often takes palpation, dermoscopy, or a biopsy, and the rules for healthy skin and tattoos need careful testing so real lesions are not missed.

While local model execution keeps patient images off cloud servers, local inference by itself does not satisfy medical privacy or device laws. Because IP geolocation, PubMed/DermNet links, Overpass queries, and map URLs all touch external servers, each module needs a clear network boundary and logging policy. Right now, the prototype is an engineering demo rather than a clinical tool.

For that reason, DermaVision is meant strictly for educational decision support: the screen warns users that model scores can be wrong and tells them to see a qualified doctor for diagnosis. Before any clinical rollout, the system would need prospective patient studies, subgroup checks across skin tones, citation-accuracy audits, and dermatologist review.

G. Evaluation Coverage and Reproducibility

Future evaluation reports should break down guard acceptance rates, per-class accuracy on accepted photos, sample counts per disorder, and exact rejection reasons. Mistakes on normal skin or tattooed skin need their own error analysis, and subgroup breakdowns should track camera model, lighting, blur, and Fitzpatrick skin type—details that are not yet in the project files.

H. Uncertainty and Module Failure Handling

Fault-tolerance checks also need to confirm that offline vision scoring still works when clinic lookup or local chat is offline, and that a failed guard check always halts disease ranking. We will need saved test logs and screenshots of the resulting user error prompts to verify both behaviors.

V. CONCLUSION

DermaVision AI brings together anatomical pre-screening, medical ViT and CLIP scoring, differential explanations, nearby clinic search, and local RAG chat in a five-stage pipeline covering 161 region–condition entries (7×23). Project records report a 92.4%

unweighted mean regional accuracy (against a 96% target), expected pass/reject results across the 237 guard test cases, and roughly 3.88 s of end-to-end CPU runtime on the test workstation. The raw dataset splits, evaluation scripts, and runtime logs are still required to verify those numbers.

Our core design goal was to filter out bad uploads early and give users transparent information they can inspect rather than a bare class label. Further study is needed to establish clinical validity, safe differential exclusion, and reliable performance across diverse patient groups. Planned next steps include prospective multicenter trials and dermoscopic video analysis on mobile devices, alongside reproducible benchmarks, calibrated uncertainty, complete regional tables, and clear offline/online boundaries.

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